

Thoughtful Labs AI Research Curriculum

A structured roadmap from LLM foundations to benchmark design and AI research.

Module 1 – LLM Foundations

Subtopics

- Transformer architecture
- Embeddings
- Tokenization
- Attention
- Feed-forward networks
- Positional encodings
- Forward pass
- Backpropagation
- Gradient descent
- Scaling intuition

Assignment

Explain how a prompt becomes the next predicted token. Draw a transformer block from memory.

Module 2 – Training vs Post-Training

Subtopics

- Pretraining
- SFT
- RLHF
- RLAIFF
- DPO
- ORPO
- GRPO
- Synthetic data
- Instruction tuning
- Continual learning

Assignment

Write a one-page comparison of pretraining vs post-training and explain where Thoughtful Labs fits.

Module 3 – Evaluations & Benchmarks

Subtopics

- Benchmark philosophy
- Prompt design
- Hidden rubrics
- Ground truth
- Pass@k
- Inter-rater reliability
- Benchmark leakage
- Difficulty calibration
- Benchmark ecosystem

Assignment

Reverse engineer five benchmark tasks and identify why each is difficult.

Module 4 – Agent Systems

Subtopics

- Planning
- Tool use
- Memory
- Retrieval
- Reflection
- Long-horizon reasoning
- Failure modes
- Evaluation

Assignment

Diagram an agent loop and explain failure modes.

Module 5 – Modern Models

Subtopics

- OpenAI
- Claude
- Gemini
- Llama
- Qwen
- DeepSeek
- Gemma
- Mistral
- MoE
- Reasoning models

Assignment

Create a comparison table for five frontier models.

Module 6 – Hugging Face

Subtopics

- Datasets
- Transformers
- TRL
- PEFT
- LoRA
- QLoRA
- Model cards
- Dataset cards
- Inference
- Evaluation

Assignment

Explore a benchmark dataset and summarize its schema.

Module 7 – Research Engineering

Subtopics

- PyTorch
- Training loops
- Losses
- Optimizers
- CUDA
- Git
- Linux
- Experiment tracking
- vLLM

Assignment

Explain every line of a simple training loop.

Module 8 – Reading Papers

Subtopics

- Paper anatomy
- Ablations
- Experimental design
- Replication
- Critical reading

Assignment

Summarize one recent LLM paper.

Module 9 – Benchmark Design

Subtopics

- Prompt writing
- Rubrics
- Atomic criteria
- Source grounding
- JSON schemas
- Difficulty calibration
- Finance benchmarks
- SEC filings
- Analyst workflows

Assignment

Build a complete benchmark task with a 100-point rubric.

Module 10 – Research Taste

Subtopics

- Research questions
- Hypothesis generation
- Experiment design
- Failure analysis
- Research writing

Assignment

Propose three novel benchmark ideas and justify them.

Capstone

Design a production-quality long-horizon benchmark for a public equity AI analyst, including a prompt, hidden 100-point rubric, supporting evidence, and validation against frontier models.